AI Prompts Being Used for Homework 1 and their Raw Responses

Prompt for Task 1: -

"You are a junior requirements engineer at a software consultancy. There is a system called ALFRED which serves as a Personal Interactive Assistant for older adults, which was developed as part of a European research project. But the product is on backlog and consists of 136 user stories written by real stakeholders. You can find the user stories in the following file - g19-alfred.txt and I am even attaching the file for your reference, here are your tasks:

1. First, identify all gaps in the requirements, especially things that are completely missing that a production system would need.
2. Next, identify all ambiguities possible, stories that are too vague to implement without further clarification and missing details.
3. Then, generate a prioritized list of clarifying questions that I can take back to stakeholders if clarification is required. Organize the questions with stakeholder roles as: OlderPerson, MedicalCaregiver, SocialCaregiver, InformalCaregiver, Developer, and System-level. Label each question with the gap or ambiguity it addresses.

Be specific in response. Do not generate generic requirement questions and try not to hallucinate. Every question must be traceable to at least one user's story in the backlog."

Raw AI response for Task 1: -

Part 1 — Gaps in the Requirements

G1 — No system administrator persona Zero stories exist for whoever deploys, configures, updates, or maintains ALFRED at an institutional or technical level. For an EU healthcare platform deployed across multiple countries, this is a critical missing stakeholder. Traceable to: Absence across all 136 stories.

G2 — No offline or connectivity failure behavior Not a single story addresses what ALFRED does when internet connectivity is lost. This is especially dangerous given that emergency contact, fall detection alerts, and caregiver communication all depend on connectivity — and many older adults live in areas with poor network coverage. Traceable to: "I want to use ALFRED to contact help in case of an emergency." / "I want to have ALFRED to detect falls and send an emergency to a specific contact."

G3 — No onboarding or offboarding process There is one story about an "introduction conversation" but nothing about how a caregiver or family member sets up a new profile, what happens when a user passes away or moves to a care facility, or how the system handles account transfer or deletion. Traceable to: "I want to have an introduction conversation with ALFRED, so that I can learn everything about him and he about me."

G4 — No app marketplace governance Several stories reference the "ALFREDO platform" and a developer marketplace, but there is no story about app review or approval, content moderation, age-appropriateness checks, what happens when a third-party app exposes user health data, or how malicious apps are removed. Traceable to: "I want to have an ALFRED device that responds to the changing demand of the users by offering new Apps in the ALFREDO platform." / "I want to easily create and to maintain my apps within the market place."

G5 — No regulatory or compliance requirements The backlog contains no stories addressing GDPR data subject rights such as access, erasure, and portability. There is no mention of EU Medical Device Regulation compliance, no data retention policy, and no audit trail requirements — all mandatory for a health-adjacent EU platform. Traceable to: "I want to know exactly what ALFRED does with my personal data, and share it only on my specific permission."

G6 — No authentication or identity verification There is no story about how ALFRED verifies who is speaking to it beyond "only reacts to me," which is a desire, not a specified mechanism. Nothing covers caregiver login, session management, or what happens if an unauthorized person attempts to access the system. Traceable to: "I want to have an ALFRED device that only reacts to me."

G7 — No software update or version management No story covers how ALFRED software updates are delivered, whether updates require user action, what happens if an update breaks existing functionality, or whether updates can be rolled back. Traceable to: Absence across all 136 stories.

G8 — No error handling or failure states No story describes what ALFRED should do when a feature fails — speech recognition errors, sensor read failures, network timeouts, app crashes — beyond the developer crash reporting stories which address only the developer's view, not the user experience of failure. Traceable to: "I want to have detailed information of what happened when my App crashes on the user's device."

G9 — No multi-user household scenario No story considers that an older person may share a home with a spouse or another older adult who also uses ALFRED. The backlog assumes a strict one-device-one-user model, which may not reflect reality. Traceable to: Absence across all 136 stories.

G10 — No payment security or transaction failure handling Two stories mention paying "small amounts" but there is no story about what happens when a payment fails, how payment credentials are stored, or what fraud protection exists. In the EU, low-value payments are governed by PSD2 which defines specific thresholds and security requirements — none of which appear in the backlog. Traceable to: "I want to use ALFRED to pay for small amounts." / "I want to be able to pay for small amounts."

G11 — No caregiver notification preference stories Multiple stories imply caregivers receive notifications about medication, activity, fall detection, and exercise performance, but no story specifies how caregivers want to receive those notifications, at what frequency, or how they manage notification load when monitoring multiple users simultaneously. Traceable to: "I want to have feedback on the activity level of the user." / "I want to be able to see if the patient

has taken her medicine." / "I want to have a regular summary on how much exercise is performed by the user."

G12 — No clinical validation of serious games and exercise features Stories about muscle group training, cognitive exercises, and health assessment games carry implicit medical claims. No story addresses whether any of these features have been clinically validated or whether making health-related claims without validation creates a regulatory liability under EU MDR. Traceable to: "I want to have some games that show me how good my memory and physical health is." / "I want to have ALFRED to offer training for specific muscle groups." / "I want to have ALFRED to propose training that promotes complex thinking."

Part 2 — Ambiguities in the Requirements

A1 — "As much as possible with speech interaction" Traceable to: "I want to use ALFRED as much as possible with speech interaction." "As much as possible" is not implementable. There is no definition of which features must be voice-accessible versus touch-only, nor any fallback hierarchy when speech fails.

A2 — "Speaks very slowly and clearly" Traceable to: "I want to have an ALFRED device that speaks very slowly and clearly." No words-per-minute target, no pronunciation clarity standard, and no specification of whether speech speed is user-adjustable or fixed at one rate.

A3 — "Only reacts to me" contradicts "push a specific button" Traceable to: "I want to have an ALFRED device that only reacts to me." / "I want to activate the vocal interaction mode of ALFRED by pushing a specific button." The first story implies passive voice recognition or biometric speaker identification. The second implies manual button activation. These are architecturally different mechanisms and the backlog never resolves which is primary.

A4 — "Everything preinstalled" Traceable to: "I want to have everything preinstalled, so that I do not have to set up ALFRED by myself." Who performs the setup instead? A family member, retailer, or caregiver? What constitutes "everything" — contacts, health profile, preferences, language? Entirely undefined.

A5 — "Very good battery" and "the whole day" Traceable to: "I want to have a very good battery for ALFRED so I can keep him on the whole day." No numeric targets for battery life. No specification of whether this applies to the phone, the wearable bracelet, or both. "The whole day" could mean 8, 12, or 16 hours.

A6 — Accessibility stories appear three times with overlapping scope Traceable to: "I want to have an ALFRED device that can be easily adapted for user that have visual or hearing troubles." / "I want to have an ALFRED device that can be easily adapted for user that have visual troubles." / "I want to have an ALFRED device that can be easily adapted for user that have hearing trouble." These three stories are likely expressing the same requirement three times. "Easily adapted" is also vague — it is unclear whether this means built-in accessibility settings, hardware variants, or third-party assistive technology compatibility.

A7 — "Pay for small amounts" appears twice Traceable to: "I want to use ALFRED to pay for small amounts." / "I want to be able to pay for small amounts." These appear to be duplicate stories. "Small amounts" is undefined — no currency, no upper limit, no payment method, and no security requirement is specified.

A8 — "Send an emergency to a specific contact" Traceable to: "I want to have ALFRED to detect falls and send an emergency to a specific contact." "Specific contact" is never defined — who sets this contact, when, and how? What is the notification method — phone call, SMS, or app alert? What happens if the contact is unreachable? The entire escalation chain is absent.

A9 — "Spontaneous questions" contradicts "programme it during certain times" Traceable to: "I want to have ALFRED ask spontaneous questions but it would be better to programme it during certain times of the day." "Spontaneous" and "programmed at certain times" are contradictory. It is also unclear who programs the schedule, what domains the questions cover, and whether the user can dismiss or silence them.

A10 — "Goals and missions" Traceable to: "I want to have ALFRED to give me goals and missions that encourage me to do new activities and exercises and to go out." No definition of what a goal or mission means in system terms, how they are generated — AI-suggested, caregiver-prescribed, or user-defined — how completion is tracked, or what happens when goals are not met.

A11 — "Data stored directly on the phone" contradicts "ALFRED on all my computers, tablets and phone" Traceable to: "I want to have my data stored directly on the phone." / "I want to have ALFRED on all my computers, tablets and phone." Local-only storage and multi-device sync are architecturally contradictory. No story resolves this tension.

A12 — "Locate the ALFRED user" Traceable to: "As a MedicalCaregiver, I want to locate the ALFRED user." Location precision is entirely undefined — city-level, neighborhood-level, or GPS-accurate. Real-time continuous tracking versus last-known-location is not specified. Legal consent requirements for tracking a vulnerable adult are unaddressed.

A13 — "Tips to reduce health impairments without medication" Traceable to: "I want to receive specific tips how to reduce my health impairments without medication." This is a medically sensitive requirement with no specification of the information source, clinical review process, liability boundaries, or which conditions and impairments are in scope.

A14 — "A few rules I have to stand to" Traceable to: "I want to be able to design my App on my own — though there will be a few rules I have to stand to." The rules are never stated anywhere in the backlog. This is a direct reference to an undefined constraint.

A15 — Malformed story: body weight Traceable to: "I want to help me to control my body weight." The story has a grammatical subject switch mid-sentence, suggesting poor transcription from a user interview. The requirement may not accurately reflect what the user actually said and needs verification with the original research team.

Part 3 — Clarifying Questions by Stakeholder, Ordered by Priority

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OlderPerson

[A8] Q1. When ALFRED detects a fall, walk me through exactly what should happen next — who gets contacted, how (phone call, text message, or app alert), in what order, and what happens if that person does not answer? Traceable to: "I want to have ALFRED to detect falls and send an emergency to a specific contact."

[A11] Q2. You mentioned wanting data stored directly on your phone, but also wanting ALFRED on all your devices. If you update your agenda on a tablet, should your phone automatically reflect that change — or should each device stay independent? Traceable to: "I want to have my data stored directly on the phone." / "I want to have ALFRED on all my computers, tablets and phone."

[A3] Q3. You said ALFRED should only react to you — how should it recognize that it is you speaking? Should it learn your voice over time, use a physical button, or both? What should happen if someone else in the room speaks to it? Traceable to: "I want to have an ALFRED device that only reacts to me." / "I want to activate the vocal interaction mode of ALFRED by pushing a specific button."

[G2] Q4. If your internet connection goes down, which features of ALFRED are most critical to keep working? Should emergency contact and fall detection function without any internet connection? Traceable to: "I want to use ALFRED to contact help in case of an emergency." / "I want to have ALFRED to detect falls and send an emergency to a specific contact."

[G3] Q5. Who currently sets up your phone or tablet for you? Would you want that same person to set up ALFRED, or would you prefer a professional from the shop where you buy it? Traceable to: "I want to have everything preinstalled, so that I do not have to set up ALFRED by myself." / "I want to buy ALFRED in a senior shop with special client support."

[A9] Q6. You said you would like ALFRED to ask you questions during the day — who should decide when those questions happen and what topics they cover? You, your caregiver, or ALFRED itself? Traceable to: "I want to have ALFRED ask spontaneous questions but it would be better to programme it during certain times of the day." / "I want to have ALFRED to initiate questions, but it should be in the domains of my interest such as cinema, exhibitions, cooking."

[A7] Q7. You mentioned using ALFRED to pay for small amounts — what do you consider a small amount? What payment method would you prefer — bank card, mobile payment, or something else? Traceable to: "I want to use ALFRED to pay for small amounts." / "I want to be able to pay for small amounts."

[A6] Q8. For users with visual trouble, which adaptations matter most — larger text, higher contrast, audio descriptions of what is on screen, or compatibility with a device you already use? Traceable to: "I want to have an ALFRED device that can be easily adapted for user that have visual or hearing troubles." / "I want to have an ALFRED device that can be easily adapted for user that have visual troubles."

[A2] Q9. When you say ALFRED should speak slowly and clearly, how slow is slow enough? Would you want to adjust the speed yourself, or should it stay at one fixed pace? Traceable to: "I want to have an ALFRED device that speaks very slowly and clearly."

[A5] Q10. When you say you want ALFRED on "the whole day," does that include overnight monitoring while you sleep, or only during waking hours? Approximately how many hours is that? Traceable to: "I want to have a very good battery for ALFRED so I can keep him on the whole day."

MedicalCaregiver

[A12] Q11. When you want to find where an ALFRED user is — do you need their exact location in real time, or would knowing where they were last seen be enough? Would you expect the user to agree to this once upfront, or each time you check? Traceable to: "As a MedicalCaregiver, I want to locate the ALFRED user."

[G5] Q12. What rules or regulations does your organization follow when accessing patient data? Are there any existing health record systems that ALFRED would need to connect with? Traceable to: "I want to know exactly what ALFRED does with my personal data, and share it only on my specific permission."

[A13] Q13. The backlog includes a story about ALFRED giving tips to reduce health impairments without medication — who should be responsible for making sure those tips are medically accurate? Should a doctor or nurse review them before they reach the user? Traceable to: "I want to receive specific tips how to reduce my health impairments without medication."

[sensor stories] Q14. You requested blood pressure, temperature, heart rate, and breathing frequency monitoring — should these run continuously in the background, or only take a reading when you or the user requests one? How accurate do the sensors need to be for you to trust the readings clinically? Traceable to: "I want to have ALFRED to be able to determine the user's blood pressure." / "I want to have ALFRED to be able to determine the user's body temperature." / "I want to have ALFRED to be able to determine the user's heart rate." / "I want to have ALFRED to be able to determine the user's breathing frequency."

[A8 + G2] Q15. If a patient's device loses internet connection and a fall is detected, what would you want to happen next? Traceable to: "I want to have ALFRED to detect falls and send an emergency to a specific contact." / "I want to use video calls in case of emergency."

[exercise stories] Q16. When exercises are performed incorrectly — does ALFRED deliver corrective feedback to the user automatically in real time, or do you first review the sensor data and then manually send a message? Traceable to: "I want to control if the user does the exercise correct." / "I want to give corrective feedback if exercises are performed wrong, too much or in any other way harmful to the user."

[G11] Q17. You will receive notifications from ALFRED about several things — medication, activity levels, exercise, and emergencies. How would you prefer to receive these — as they happen, as a daily summary, or as a dashboard you check yourself? What should trigger an immediate alert versus a routine update? Traceable to: "I want to have feedback on the activity

level of the user." / "I want to be able to see if the patient has taken her medicine." / "I want to have a regular summary on how much exercise is performed by the user."

SocialCaregiver

[G — role definition] Q18. The backlog has only four stories for SocialCaregiver. Can you describe your full role — what information do you need access to, and what actions can you take that a MedicalCaregiver cannot, and vice versa? Traceable to: "I want ALFRED to remind me of the things I need to bring to the person I care for." / "I want to have ALFRED to be able to recognize pain and or stress by using skin conduction sensors." / "I want to have a step counter included into ALFRED."

[skin conduction story] Q19. When stress or pain is detected via skin conduction sensors, what should ALFRED do — alert you immediately, log it silently for later review, or ask the user directly before escalating? Traceable to: "I want to have ALFRED to be able to recognize pain and or stress by using skin conduction sensors."

[step counter story] Q20. For the step counter — is a daily total sufficient, or do you need hourly breakdowns, trend analysis over weeks, or comparison against a prescribed daily target? Traceable to: "I want to have a step counter included into ALFRED."

InformalCaregiver

[G — role definition] Q21. There is only one story for InformalCaregiver in the entire backlog. Can you tell me more about your role — are you mainly there for social contact and shared activities, or do you also need to see health information? How is your role different from a SocialCaregiver's? Traceable to: "I want to have a game that the carer could play together with the user, so that it would enable to monitor the users' state of health but also have some fun time."

[shared game story] Q22. The story about playing a game together with the user — does this need to work when you are not physically there, or only when you are in the same room? Should any health information be shared with you during the game, or is that something the user would decide? Traceable to: "I want to have a game that the carer could play together with the user, so that it would enable to monitor the users' state of health but also have some fun time."

Developer

[A14] Q23. The backlog mentions that developers must follow a few rules when designing apps — what are those rules exactly? Is there a content policy, a prohibited categories list, a review process, or something else before an app can be published? Traceable to: "I want to be able to design my App on my own — though there will be a few rules I have to stand to."

[G4] Q24. If a third-party app accesses health sensor data and suffers a data breach, who is legally responsible — the app developer, the ALFRED platform, or the healthcare organization that deployed it? Traceable to: "I want to have an easy way to read data from health sensors the user is wearing." / "I want to easily create and to maintain my apps within the market place."

[OS access story] Q25. You want apps to access several operating system features — which ones specifically? Would the older person need to individually approve each app's access to each feature? Traceable to: "I want to have my App to be granted access to several operating system features."

[pricing story] Q26. The backlog asks what fees the marketplace charges but never answers it. What is the intended revenue model — a percentage of sales, a flat listing fee, or free for all developers? Traceable to: "I want to know the fees the provider of the market place charges for using the platform."

[crash reporting story] Q27. When an app crashes, what information should be captured and sent to you — what the user was doing, the device state, the error itself? Would any of that information identify the user personally, and if so how should it be handled? Traceable to: "I want to have detailed information of what happened when my App crashes on the user's device."

[G13] Q28. The final two stories in the backlog are word-for-word identical — "I want to have the ALFRED API to reduce the work to achieve my goals." Was this intentional to signal high priority, or is it a data entry error that should be removed? Traceable to: Final two entries in g19-alfred.txt.

System-level

[G5] Q29. How should ALFRED be classified under EU law — as a medical device, a general wellness product, or a social assistance tool? The answer to this determines what rules the entire system must follow before it can be deployed. Traceable to: "I want to know exactly what ALFRED does with my personal data, and share it only on my specific permission." / "I want to receive specific tips how to reduce my health impairments without medication."

[G6] Q30. How does ALFRED verify the identity of a MedicalCaregiver or SocialCaregiver accessing a user's data? Is there a login system, different access levels per role, or a two-step verification process? Traceable to: "I want to decide who has access to my vital data." / "I want to be sure that the user can decide who has access to her data."

[G9] Q31. What happens if two people in the same household both use ALFRED? Can one device serve two users, or does each person need their own? How does ALFRED tell two voices apart when both people are in the same room? Traceable to: "I want to have an ALFRED device that only reacts to me." / "I want to have an ALFRED device that only listens to me when I ask it to."

[G7] Q32. How are software updates delivered to ALFRED devices? Are they automatic, or does the user or caregiver need to approve them? What happens if an update fails on a device

with no technical support nearby? Traceable to: Absence across all 136 stories — identified as a production system gap.

[G8] Q33. What should ALFRED do when it repeatedly cannot understand what the user is saying? After how many failed attempts should it switch to a different way of interacting — and what should that be? Traceable to: "I want to use ALFRED as much as possible with speech interaction." / "I want to have an ALFRED device that speaks very slowly and clearly."

[G3] Q34. What is the process for shutting down ALFRED when a user no longer needs it — for example if they move into full-time care? Who starts that process, what gets deleted, and what gets kept and for how long? Traceable to: "I want to have an introduction conversation with ALFRED, so that I can learn everything about him and he about me." / "I want to have everything preinstalled, so that I do not have to set up ALFRED by myself."

[G12] Q35. Several stories mention exercise and health games that assess memory and physical condition. Has any of this content been reviewed or approved by a medical professional? Are there any clinical guidelines the exercise and games features must follow? Traceable to: "I want to have some games that show me how good my memory and physical health is." / "I want to have ALFRED to offer training for specific muscle groups." / "I want to have ALFRED to propose training that promotes complex thinking."

Prompt for Task 2: - "For the same dataset g19-alfred.txt could you classify and extract requirements from it into the following four categories: Functional Requirements (FR), Non-Functional Requirements (NFR) with subcategories such as usability, security, performance, portability, and reliability, Business Rules, and Constraints. Be specific and do not assume. Every requirement you extract must be traceable to user story in the backlog. Do not invent requirements that are not present in the stories."

Raw AI response for Task 2: -

Functional Requirements (FR)

FR1 — Speech & Interaction

FR1.1 The system shall deliver information to the user via speech technology.

FR1.2 The system shall support speech as the primary interaction mode.

FR1.3 The system shall allow the user to activate voice interaction mode by pressing a specific physical button.

FR1.4 The system shall repeat the last spoken phrase on user request.

FR1.5 The system shall support speech input for inserting meetings and agenda entries.

FR1.6 The system shall conduct an introductory conversation with the user during first setup to learn user preferences.

FR1.7 The system shall only listen to the user when explicitly asked to do so.

FR1.8 The system shall only react to the voice of its registered user.

FR1.9 The system shall allow developers to define questions the user can ask and perform via the API.

FR2 — Health Monitoring & Vitals

FR2.1 The system shall monitor and report the user's blood pressure.

FR2.2 The system shall monitor and report the user's body temperature.

FR2.3 The system shall monitor and report the user's heart rate.

FR2.4 The system shall monitor and report the user's breathing frequency.

FR2.5 The system shall provide the MedicalCaregiver with feedback on the user's activity level.

FR2.6 The system shall allow the user to view their own vital parameters on the phone.

FR2.7 The system shall capture health data using sensors while the user performs guided exercises.

FR2.8 The system shall recognize pain or stress using skin conduction sensors.

FR2.9 The system shall include a step counter.

FR2.10 The system shall allow the MedicalCaregiver to locate the ALFRED user.

FR2.11 The system shall support body weight monitoring for the user.

FR3 — Medication Management

FR3.1 The system shall send medicine reminders to the user.

FR3.2 The system shall allow the user to check whether they have already taken their medication to avoid overdose.

FR3.3 The system shall allow the MedicalCaregiver to check whether the patient has taken their medication.

FR4 — Emergency & Safety

FR4.1 The system shall detect falls and send an emergency alert to a specific pre-configured contact.

FR4.2 The system shall allow the user to contact help in case of an emergency.

FR4.3 The system shall contact the nearest caregiver when the user explicitly requests urgent help.

FR4.4 The system shall allow the user to call their caregiver by pressing a single button.

FR4.5 The system shall support video calls for MedicalCaregivers in emergency situations.

FR4.6 The system shall notify the user when their home care nurse is on the way.

FR5 — Agenda, Reminders & Daily Routine

FR5.1 The system shall manage the user's agenda.

FR5.2 The system shall send reminders for appointments.

FR5.3 The system shall provide day and night rhythm support with agenda reminders.

FR5.4 The system shall send a morning wake-up call and check on the user's wellbeing.

FR5.5 The system shall remind the user of items to bring when leaving the house.

FR5.6 The system shall remind the SocialCaregiver of items to bring when visiting the user.

FR5.7 The system shall initiate proactive questions to the user in domains of their interest at programmable times of day.

FR5.8 The system shall notify the user of low battery status before they leave the house.

FR6 — Communication & Social

FR6.1 The system shall allow the user to make phone calls.

FR6.2 The system shall allow the user to send messages.

FR6.3 The system shall support video conversations.

FR6.4 The system shall provide videophone capability for MedicalCaregivers.

FR6.5 The system shall maintain the user's contact list.

FR6.6 The system shall allow the user to contact people from their own social circle.

FR6.7 The system shall allow the user to set up social groups of people with similar diseases.

FR6.8 The system shall allow the user to organize group meetings with friends at a specific place.

FR6.9 The system shall encourage the user toward face-to-face human interaction rather than only digital contact.

FR7 — Social Activities & Events

FR7.1 The system shall inform the user about art expositions and museums.

FR7.2 The system shall inform the user about social activities organized in their neighborhood.

FR7.3 The system shall send personalized invitations to social events to motivate the user to attend.

FR7.4 The system shall suggest sports classes available in the user's neighborhood.

FR7.5 The system shall provide information about existing apps that correspond to the user's interests.

FR8 — Navigation & Transportation

FR8.1 The system shall provide navigation assistance to the user.

FR8.2 The system shall support navigation to events and concerts and assist the user when lost.

FR8.3 The system shall support navigation for users with mild cognitive impairments.

FR8.4 The system shall provide public transport usage support.

FR8.5 The system shall assist the user in arranging a taxi on time.

FR8.6 The system shall provide touristic walk guidance with explanations of historical and cultural sights.

FR9 — Smart Home Control

FR9.1 The system shall allow the user to see who is at the door and remotely open it.

FR9.2 The system shall allow the user to turn lights on and off.

FR9.3 The system shall allow the user to control the heating.

FR9.4 The system shall allow the user to control the TV and radio.

FR9.5 The system shall allow the user to answer incoming phone calls more quickly when at home.

FR10 — Entertainment & Information

FR10.1 The system shall allow the user to watch movies.

FR10.2 The system shall provide news to the user.

FR10.3 The system shall suggest healthy recipes to the user.

FR10.4 The system shall allow the user to do grocery shopping.

FR11 — Exercise & Serious Games

FR11.1 The system shall provide guided exercises to the user.

FR11.2 The system shall provide games that combine mental and physical exercise such as opposing motion exercises.

FR11.3 The system shall provide games that assess memory and physical health.

FR11.4 The system shall provide training for specific muscle groups.

FR11.5 The system shall provide games that include the user's favorite sports.

FR11.6 The system shall support online multiplayer games with other users.

FR11.7 The system shall support a shared game that an InformalCaregiver and user can play together to monitor the user's health state.

FR11.8 The system shall allow the user to play imagination-based games.

FR11.9 The system shall set time limits on game sessions to prevent user boredom.

FR11.10 The system shall assign goals and missions to encourage the user to do new activities and go out.

FR11.11 The system shall remind the user to maintain good body posture.

FR11.12 The system shall remind users to exercise regularly.

FR11.13 The system shall vary exercise recommendations daily to maintain user motivation.

FR11.14 The system shall allow the user to view their training performance.

FR11.15 The system shall propose training that promotes complex thinking.

FR11.16 The system shall allow the MedicalCaregiver to verify whether exercises are performed correctly.

FR11.17 The system shall provide the MedicalCaregiver with a regular summary of exercise performed by the user.

FR11.18 The system shall allow the serious games developer to take control over specific system functions to provide the desired user experience.

FR11.19 The system shall allow the MedicalCaregiver to give corrective feedback when exercises are performed wrong, too much, or in a harmful way.

FR12 — Privacy & Data Control

FR12.1 The system shall inform the user exactly what it does with their personal data.

FR12.2 The system shall share personal data only with the user's explicit permission.

FR12.3 The system shall allow the user to manage their privacy and data settings.

FR12.4 The system shall allow the user to decide who has access to their vital data.

FR12.5 The system shall ensure the user retains full control over all ALFRED activities.

FR13 — Developer Platform & API

FR13.1 The system shall provide a well-defined and well-documented API.

FR13.2 The system shall provide high-level API components that abstract low-level implementation details.

FR13.3 The system shall provide an API that reduces developer effort to achieve their goals.

FR13.4 The system shall provide an easy way to read data from health sensors the user is wearing.

FR13.5 The system shall allow developers to easily create and maintain apps within the marketplace.

FR13.6 The system shall allow developers to manage pricing modalities for their apps.

FR13.7 The system shall allow developers to present their app in multiple formats or views.

FR13.8 The system shall allow developers to collect and view user feedback on their apps.

FR13.9 The system shall notify developers as soon as the status of any of their apps changes.

FR13.10 The system shall provide developers with detailed crash information when their app fails on a user's device.

FR13.11 The system shall provide developers with simple and clear accounting for their apps.

FR13.12 The system shall provide developers with app development support.

FR13.13 The system shall grant apps access to specified operating system features.

FR13.14 The system shall provide an easy way for developers to manage user settings.

FR14 — Payments

FR14.1 The system shall allow the user to make payments for small amounts.

Non-Functional Requirements (NFR)

NFR1 — Usability

NFR1.1 The system shall display only well-visible large buttons suitable for older adults.

NFR1.2 The system shall provide visual support when displaying lists, maps, and agendas.

NFR1.3 The system shall allow the user to zoom into all visual content.

NFR1.4 The system shall speak at a slow and clear pace appropriate for older adults.

NFR1.5 The system shall be adaptable for users with visual impairments.

NFR1.6 The system shall be adaptable for users with hearing impairments.

NFR1.7 The system shall allow the user to customize preferences across all functionalities.

NFR1.8 The system shall come fully preinstalled and configured so the user does not need to perform any setup.

NFR1.9 The system shall provide clear instructions and post-sales support for users who experience difficulty.

NFR1.10 The system shall be available across all of the user's computers, tablets, and phones.

NFR1.11 The system shall support a simple expandable unit structure for MedicalCaregivers.

NFR2 — Security & Privacy

NFR2.1 The system shall ensure that personal data is shared only with the user's explicit permission.

NFR2.2 The system shall ensure the MedicalCaregiver is confident that the user retains control over who accesses their data.

NFR2.3 The system shall store user data directly on the user's phone.

NFR3 — Performance

NFR3.1 The system shall maintain sufficient battery life to operate throughout the entire day without recharging.

NFR3.2 The system shall deliver developer app status notifications in near real-time.

NFR4 — Portability & Device

NFR4.1 The system shall be available in a small wearable form factor that is easy to carry.

NFR4.2 The system shall be available in a bracelet or necklace form factor.

NFR4.3 The system shall be available as a wearable shirt with integrated sensors.

NFR5 — Aesthetics & Social Acceptability

NFR5.1 The wearable device with integrated sensors shall have an appearance that is not stigmatizing.

NFR5.2 The ALFRED shirt shall look good and be non-stigmatizing.

NFR6 — Extensibility & Maintainability

NFR6.1 The system shall respond to changing user demands by offering new apps through the ALFREDO platform.

NFR6.2 The developer API shall be designed at a high level of abstraction to minimize developer effort.

NFR7 — Reliability & Safety

NFR7.1 Health tips provided by ALFRED regarding reducing impairments without medication must not conflict with professional medical advice.

Business Rules (BR)

BR1 Personal data shall only be shared with third parties when the user has given explicit specific permission. Blanket consent is not sufficient.

BR2 The user shall always retain the right to decide who has access to their vital health data including caregivers.

BR3 App developers must adhere to a defined set of platform rules when designing and publishing apps on the ALFREDO marketplace. The specific rules are undefined in the backlog and must be elicited separately.

BR4 The marketplace operator charges fees to developers for using the platform. The fee structure must be transparent and known to developers before they publish.

BR5 Developers shall be able to define their own app pricing modality within the marketplace.

BR6 Emergency alerts triggered by fall detection must be sent to a pre-configured specific contact and not broadcast indiscriminately.

BR7 ALFRED shall contact the nearest available caregiver and not a fixed single contact when the user explicitly requests urgent help.

BR8 The system shall encourage face-to-face human interaction and shall not position itself as a replacement for real human contact.

BR9 Health tips provided by ALFRED regarding reducing impairments without medication must not substitute professional medical advice.

BR10 Corrective exercise feedback from the MedicalCaregiver shall only be delivered if exercises are performed wrongly, excessively, or in a way harmful to the user and not as routine commentary.

Constraints (CO)

CO1 ALFRED must be purchasable through senior shops with dedicated client support, constraining distribution to specialized retail channels only.

CO2 ALFRED must be available at a low price point accessible to older adults, constraining hardware and licensing cost decisions.

CO3 User data must be stored directly on the user's phone. Cloud-only storage is not acceptable. This directly conflicts with the multi-device availability requirement in NFR1.10 and must be resolved during architecture design.

CO4 The wearable device must be small enough to be worn as a bracelet or necklace at all times, constraining hardware size, weight, and battery integration.

CO5 All wearable form factors including the shirt and bracelet must not resemble medical devices, constraining industrial design decisions.

CO6 Third-party developers are constrained by platform rules defined by the ALFRED marketplace operator. Apps that do not comply cannot be published. The actual rules are undefined in the backlog and must be specified before the marketplace can be built.

CO7 The system must arrive fully configured and ready to use. No user-facing setup steps are permitted.

CO8 Payment functionality is explicitly limited to small amounts only. Large-value transactions are out of scope.

CO9 The system must not propose or continue exercises that are being performed in a harmful way. Exercise recommendations are bounded by user safety limits set by the MedicalCaregiver.

CO10 The ALFRED API must abstract all low-level system and sensor details from developers. Developers must not be required to interact directly with hardware-level components.

Prompt for Task 3: - "Using the requirements we have extracted in Task 2, you need to perform two more tasks. First go through them and identify any conflicts or tensions between generated requirements, i.e. in case wherever two requirements pull the system in opposite directions or cannot both be fully satisfied at the same time, tell me which specific requirement IDs are clashing, what the tension is about, and what goes wrong at implementation if it is not resolved.

Then once you have done that, take the full list of Functional, Non-Functional Requirements, Business Rules, and Constraints and prioritize them using MOSCOW criteria, like which of them are "Must Have", "Should Have", "Could Have", and "Won't Have". Also provide a short justification for each, keeping in mind that the anchor for what counts as Must Have is whatever is essential for a safe, legally compliant, and working minimum viable product for older adults in a European healthcare context. Do not invent any new conflicts or priorities, everything must be directly traceable to the requirements list created in task 2**."**

Raw AI response for Task 3: -

Part A — Conflict Detection

Conflict 1 — Local data storage vs. multi-device availability Requirements: NFR2.3 / CO3 vs. NFR1.10

NFR2.3 and CO3 both require user data to be stored directly on the phone. NFR1.10 requires ALFRED to be available across all of the user's computers, tablets, and phones. These are architecturally incompatible. If data lives only on the phone, a tablet cannot access it without network sync. If data syncs across devices, it cannot be stored locally only. If left unresolved, the development team cannot make a fundamental architecture decision — any choice they make will violate one of the two requirements.

Conflict 2 — User privacy and data control vs. caregiver monitoring Requirements: FR12.2, FR12.4, BR1, BR2 vs. FR2.5, FR2.10, FR3.3, FR11.16, FR11.17

The user must share data only on explicit permission and must decide who accesses their vital data. At the same time MedicalCaregivers need continuous feedback on activity levels, location access, medication adherence tracking, and exercise performance summaries. In a real emergency or cognitive decline scenario the user may be unable or unwilling to grant permission in the moment. If left unresolved, either caregiver safety monitoring stops when the user withholds consent, or the system violates its own privacy requirements by overriding the user's decision.

Conflict 3 — Autonomous fall detection vs. user control over ALFRED Requirements: FR4.1, FR4.3 vs. FR12.5

FR4.1 requires ALFRED to autonomously detect falls and send emergency alerts without user instruction. FR12.5 requires the user to always retain full control over all ALFRED activities. Autonomous fall detection by definition acts without the user triggering it — it fires automatically. If taken literally FR12.5 would prevent autonomous emergency responses, making the safety features non-functional. If left unresolved, either the safety feature is disabled to preserve user control, or user control is silently overridden to preserve safety.

Conflict 4 — Proactive questioning vs. user control and alert fatigue Requirements: FR5.7 vs. FR12.5, NFR1.7

FR5.7 requires ALFRED to initiate proactive questions at programmable times of day. FR12.5 requires the user to always retain full control. NFR1.7 requires the user to customize preferences across all functionalities. If ALFRED initiates questions autonomously it is taking an action the user did not trigger in that moment, conflicting with full user control. Additionally FR5.7 combined with FR3.1, FR5.2, FR5.4, FR11.11, and FR11.12 creates a high cumulative notification load. If left unresolved the system risks alert fatigue, causing users to ignore or disable reminders including safety-critical ones like medication and emergency alerts.

Conflict 5 — Rich health data collection vs. data minimization and informed consent Requirements: FR2.1, FR2.2, FR2.3, FR2.4, FR2.7, FR2.8, FR2.9 vs. FR12.1, FR12.2, BR1, NFR2.1

The system collects blood pressure, temperature, heart rate, breathing frequency, skin conduction for pain and stress, step count, and exercise sensor data continuously. At the same time requirements mandate that data is only shared on explicit permission and that the user is informed of everything ALFRED does with their data. Continuous passive sensor collection creates a tension with informed consent — the user cannot be asked for permission before every individual sensor reading. If left unresolved a blanket upfront consent may satisfy the letter of the requirement but not its spirit, creating a legal compliance risk under GDPR.

Conflict 6 — Low price constraint vs. comprehensive hardware requirements Requirements: CO2 vs. CO4, NFR3.1, NFR4.1, NFR4.2, NFR4.3, FR2.1, FR2.2, FR2.3, FR2.4, FR2.8, FR4.1

CO2 requires ALFRED to be available at a low price accessible to older adults on fixed incomes. The hardware requirements simultaneously demand a bracelet or necklace wearable, a sensor-equipped shirt, blood pressure monitoring, skin conduction sensors, step counter, fall detection, and all-day battery life. Delivering all of this at a low price point is a direct commercial conflict. If left unresolved the product either ships incomplete or ships at a price that excludes the very users it is designed for.

Conflict 7 — Non-stigmatizing wearable design vs. comprehensive sensor integration Requirements: NFR5.1, NFR5.2, CO5 vs. FR2.1, FR2.2, FR2.3, FR2.4, FR2.8, FR4.1

The system must not look like a medical device and must look good and be non-stigmatizing. At the same time it must measure blood pressure, body temperature, heart rate, breathing frequency, skin conduction for stress and pain, and detect falls. Current sensor technology for clinical-grade vital monitoring is bulky and typically visible. If left unresolved the hardware design team faces an impossible brief — clinical accuracy and consumer aesthetics cannot both be fully satisfied with current wearable technology.

Conflict 8 — Developer freedom vs. undefined platform governance Requirements: FR13.5, FR13.6, FR13.7 vs. BR3, CO6

Developers want to easily create apps, manage their own pricing, and present apps in multiple ways. BR3 and CO6 constrain developers to undefined platform rules. The conflict is that the rules constraining developer freedom are never specified anywhere in the requirements list. If left unresolved developers cannot build compliant apps because they do not know what compliance means, and the marketplace cannot launch without a defined governance framework.

Conflict 9 — Encouraging face-to-face contact vs. comprehensive digital social services Requirements: FR6.9, BR8 vs. FR6.1, FR6.2, FR6.3, FR6.6, FR6.7, FR6.8, FR7.3, FR11.6

The system is required to encourage real human interaction and not position itself as a replacement for human contact. At the same time it provides video calls, messaging, social group formation, event invitations, and online multiplayer games. The more capable and convenient the digital social features become, the more they risk becoming a substitute for the face-to-face contact the system is supposed to encourage. If left unresolved no developer can determine when a social feature crosses the line from supportive to substitutive.

Conflict 10 — Daily exercise variety vs. MedicalCaregiver safety oversight Requirements: FR11.13 vs. FR11.16, FR11.19

FR11.13 requires exercise recommendations to change daily to maintain user motivation. FR11.16 requires the MedicalCaregiver to verify that exercises are performed correctly. FR11.19 requires corrective feedback when exercises are harmful. If exercises change daily without caregiver pre-approval, the caregiver cannot verify safety before the user performs a new exercise type. If left unresolved a new exercise introduced autonomously by the system may be contraindicated for a specific user's medical condition, creating a direct patient safety risk.

Conflict 11 — Disease-based social grouping vs. health data privacy Requirements: FR6.7 vs. FR12.2, BR1

FR6.7 allows users to set up social groups of people with similar diseases. FR12.2 and BR1 require that personal data is only shared with explicit permission. Joining or creating a disease-based group implicitly discloses a health diagnosis to other group members — a form of sensitive health data sharing. No requirement addresses how this disclosure is governed, whether group membership is visible to others, or whether joining constitutes explicit consent to that disclosure. If left unresolved the feature creates a GDPR-sensitive data exposure risk.

Conflict 12 — Proactive questioning vs. cognitive safety for vulnerable users Requirements: FR5.7 vs. FR12.5, target user context

FR5.7 requires ALFRED to initiate proactive questions at scheduled times. The backlog explicitly acknowledges that some users have mild cognitive impairments. For older adults with dementia or cognitive decline, unsolicited questions from a device can cause confusion, disorientation, or distress — a clinical risk that goes beyond the usability concern of alert fatigue. No requirement defines how the proactive questioning feature should be adjusted or disabled for users with cognitive impairments. If left unresolved the feature may cause direct psychological harm to a subset of the target user population.

Conflict 13 — Notification volume vs. safety-critical alert visibility Requirements: FR3.1, FR5.2, FR5.4, FR5.7, FR11.11, FR11.12 vs. FR4.1, FR4.2

Six separate requirement categories independently generate notifications — medicine reminders, appointment reminders, morning check-ins, proactive questions, posture reminders, and exercise reminders. When all of these run simultaneously alongside emergency alerts there is no requirement that defines a notification priority hierarchy. If left unresolved a safety-critical fall alert or emergency notification may be dismissed by a user who has learned to ignore the constant stream of routine ALFRED notifications, creating a direct patient safety risk.

Part B — MoSCoW Prioritization

Must Have

FR3.1 — Medicine reminders to the user. A missed medication dose for an older adult can be life-threatening. Non-negotiable.

FR3.2 — User medication self-check to avoid overdose. Directly prevents patient harm. Must be present at launch.

FR3.3 — MedicalCaregiver medication adherence check. Clinical oversight of medication compliance is a core healthcare requirement.

FR4.1 — Fall detection and emergency alert. Falls are the leading cause of injury in older adults. Core safety feature.

FR4.2 — Emergency help contact. Basic safety requirement. Cannot ship without this.

FR4.3 — Nearest caregiver contact on urgent request. Direct user safety mechanism explicitly requested by stakeholders.

FR4.4 — One-button caregiver call. Simplest possible emergency access for the target user group.

FR4.6 — Nurse on-the-way notification. Reduces user anxiety and supports care coordination.

FR1.1 — Information delivery via speech. Core interaction modality. Without this ALFRED cannot function for the target demographic.

FR1.2 — Speech as primary interaction mode. Foundational to the entire system design.

FR1.7 — Listen only when asked. Privacy-critical. A device that listens continuously without consent violates GDPR.

FR1.8 — React only to registered user. Security and privacy requirement. Prevents unauthorized access.

FR12.1 — Inform user what ALFRED does with their data. Legally required under GDPR Article 13.

FR12.2 — Share data only on explicit permission. Legally required under GDPR. Non-negotiable.

FR12.3 — User manages privacy and data settings. GDPR right to control. Must be present at launch.

FR12.4 — User decides who accesses vital data. Core privacy right directly requested by both OlderPerson and MedicalCaregiver.

FR12.5 — User retains full control over ALFRED activities. Foundational user autonomy requirement.

FR2.1 — Blood pressure monitoring. Clinically requested by MedicalCaregiver. Core health monitoring function.

FR2.2 — Body temperature monitoring. Clinically requested by MedicalCaregiver.

FR2.3 — Heart rate monitoring. Clinically requested by MedicalCaregiver.

FR2.4 — Breathing frequency monitoring. Clinically requested by MedicalCaregiver.

FR2.6 — User views own vital parameters. Supports user autonomy and self-monitoring. Directly requested.

NFR1.1 — Well-visible large buttons. Without this the interface is unusable for the target demographic.

NFR1.4 — Slow and clear speech pace. Core usability for older adults. Directly requested.

NFR1.5 — Adaptable for visual impairments. Legally required under EU European Accessibility Act.

NFR1.6 — Adaptable for hearing impairments. Legally required under EU European Accessibility Act.

NFR1.8 — Fully preinstalled and configured. Without this the user cannot set up the device independently.

NFR1.9 — Clear instructions and post-sales support. Essential for a non-technical user demographic.

NFR2.1 — Data shared only with explicit permission. GDPR legal requirement.

NFR3.1 — All-day battery life. Without this safety features like fall detection and emergency contact fail mid-day.

BR1 — Explicit specific consent required for data sharing. GDPR legal requirement.

BR2 — User retains right to control vital data access. GDPR and fundamental rights requirement.

BR6 — Emergency alerts sent to pre-configured contact only. Prevents indiscriminate data disclosure during emergencies.

BR7 — Nearest caregiver contacted on urgent help request. Directly supports emergency response.

BR9 — Health tips must not substitute medical advice. Legal liability requirement. Must be enforced from day one.

CO8 — System arrives fully configured, no user setup required. Without this the system cannot be used by the target demographic independently.

Should Have

FR1.3 — Activate voice mode via physical button. Useful fallback but not the only interaction mechanism.

FR1.4 — Repeat last spoken phrase. Important accessibility feature but system functions without it.

FR1.5 — Speech input for agenda entries. Enhances usability but agenda can be managed without speech input.

FR1.6 — Introductory setup conversation. Improves onboarding experience but not safety-critical.

FR2.5 — MedicalCaregiver activity level feedback. Important for ongoing care but not an immediate safety feature.

FR2.9 — Step counter. Useful health indicator but not clinically critical at MVP.

FR5.1 — Agenda management. Important for daily routine support.

FR5.2 — Appointment reminders. Directly supports medication and care appointments.

FR5.3 — Day and night rhythm support. Important for older adults with cognitive impairment.

FR5.4 — Morning wake-up call and wellbeing check. Important daily safety check.

FR5.5 — Reminder of items to bring when leaving house. Directly requested and practical.

FR5.8 — Low battery notification before leaving house. Important for maintaining safety feature availability.

FR6.1 — Phone calls. Core communication feature directly requested.

FR6.2 — Send messages. Core communication feature directly requested.

FR6.5 — Contact list maintenance. Supports communication features.

FR8.3 — Navigation for mild cognitive impairments. Promoted from Could Have during human review. Navigation support for cognitively impaired users is a core clinical differentiator of ALFRED versus a generic smartphone assistant.

FR11.1 — Guided exercises. Important for health maintenance.

FR11.11 — Body posture reminders. Useful health feature.

FR11.12 — Exercise reminders. Supports health maintenance.

FR11.14 — Training performance view. Supports user motivation and self-monitoring.

FR11.16 — MedicalCaregiver exercise correctness verification. Important safety feature for exercise domain.

FR11.17 — MedicalCaregiver exercise summary. Supports clinical oversight.

FR11.19 — MedicalCaregiver corrective exercise feedback. Important safety feature for exercise domain.

FR13.1 — Well-defined and documented API. Promoted from Could Have during human review. Without a documented API no third-party apps can be built and the ALFREDO marketplace cannot function.

FR13.2 — High-level API abstraction. Promoted from Could Have during human review. Directly enables the developer ecosystem the platform depends on.

FR13.4 — Easy health sensor data reading via API. Promoted from Could Have during human review. Without this third-party health and exercise apps cannot access sensor data, blocking a core product capability.

NFR1.2 — Visual support for lists, maps, agendas. Important usability feature for target demographic.

NFR1.3 — Zoom into visual content. Accessibility feature.

NFR1.7 — User customizes preferences. Important for personalization.

NFR1.10 — Available across all user devices. Important but conflicts with CO3 and needs architecture resolution first.

NFR1.11 — Expandable unit for MedicalCaregivers. Important for clinical scalability.

NFR2.2 — MedicalCaregiver confident user controls data access. Important trust requirement.

BR10 — Corrective feedback only when exercise is harmful. Important safety boundary for exercise features.

CO4 — Wearable bracelet or necklace form factor. Important for user acceptance.

CO9 — No harmful exercises permitted. Important safety constraint for exercise domain.

Could Have

FR2.7 — Sensor data capture during exercises. **FR2.8** — Pain and stress recognition via skin conduction. **FR2.10** — MedicalCaregiver user location. Moved from Should Have during human review. Has an unresolved legal consent conflict with FR12.4 and BR2. Cannot be implemented as-is until the consent framework is defined.

FR2.11 — Body weight monitoring. **FR5.6** — SocialCaregiver reminder of items to bring. **FR5.7** — Proactive interest-based questions at scheduled times. Carries alert fatigue risk (Conflict 4) and cognitive safety risk (Conflict 12). Requires additional safeguards before implementation.

FR6.3 — Video conversations. **FR6.4** — MedicalCaregiver videophone. **FR6.6** — Contact people from own social circle. **FR6.7** — Social groups by similar diseases. Carries unresolved GDPR conflict (Conflict 11). Requires consent framework before implementation.

FR6.8 — Organize group meetings. **FR7.1** — Art exposition and museum information. **FR7.2** — Neighborhood social activity information. **FR7.3** — Personalized social event invitations. **FR7.4** — Neighborhood sports class suggestions. **FR7.5** — App recommendations matching user interests. **FR8.1** — Navigation assistance. **FR8.2** — Navigation to events. **FR8.4** — Public transport support. **FR8.5** — Taxi arrangement support. **FR8.6** — Touristic walk guidance. **FR9.1** — Door camera and remote door open. **FR9.2** — Light control. **FR9.3** — Heating control. **FR9.4** — TV and radio control. **FR9.5** — Faster phone answering at home. **FR10.1** — Watch movies. **FR10.2** — News delivery. **FR10.3** — Healthy recipe suggestions. **FR10.4** — Grocery shopping. **FR11.2** — Mental and physical combined exercise games. **FR11.3** — Memory and physical health assessment games. **FR11.4** — Specific muscle group

training. **FR11.5** — Favorite sports games. **FR11.6** — Online multiplayer games. **FR11.7** — Shared InformalCaregiver and user game. **FR11.8** — Imagination-based games. **FR11.9** — Game session time limits. **FR11.10** — Goals and missions for new activities. **FR11.13** — Daily varying exercise recommendations. Carries unresolved safety conflict with FR11.16 and FR11.19 (Conflict 10). Requires caregiver approval workflow before implementation.

FR11.15 — Complex thinking training. **FR11.18** — Serious games developer system control. **FR13.3** — API reduces developer effort. **FR13.5** — Developer app creation and maintenance. **FR13.6** — Developer pricing management. **FR13.7** — App presentation in multiple formats. **FR13.8** — Developer user feedback collection. **FR13.9** — Developer app status notifications. **FR13.10** — Developer crash reporting. **FR13.11** — Developer accounting. **FR13.12** — Developer support. **FR13.13** — App OS feature access. **FR13.14** — Developer user settings management. **NFR4.1** — Small wearable form factor. **NFR4.2** — Bracelet or necklace form factor. **NFR4.3** — Wearable shirt with sensors. **NFR5.1** — Non-stigmatizing wearable appearance. **NFR5.2** — Non-stigmatizing shirt appearance. **NFR6.1** — New apps offered through ALFREDO platform. **NFR6.2** — High-level API abstraction for developers. **BR3** — Developer platform rules compliance. **BR4** — Transparent marketplace fee structure. **BR5** — Developer controls own pricing. **BR8** — System encourages face-to-face interaction. **CO1** — Distribution through senior shops only. **CO6** — Developer platform rules constraint. **CO11** — API abstracts hardware from developers.

Won't Have

FR14.1 — Payments for small amounts. Payment security, fraud protection, transaction failure handling, and PSD2 regulatory compliance are entirely absent from the requirements list. Implementing payments without these creates legal and security risk.

CO2 — Low price constraint as currently stated. Directly conflicts with CO4, NFR3.1, NFR4.1, NFR4.2, NFR4.3, and FR2.1 through FR2.4. The full hardware specification cannot be delivered at a consumer-accessible low price. Requires renegotiation before it can be treated as a fixed constraint.

CO3 as an absolute constraint — Local-only data storage. Makes NFR1.10 multi-device availability impossible to implement. Must be resolved architecturally before being enforced as a hard constraint.

BR3 and CO7 as enforceable constraints — Developer platform rules. The actual rules are undefined anywhere in the requirements list. Developers cannot comply with rules that do not exist. These become enforceable only after the rules themselves are specified.